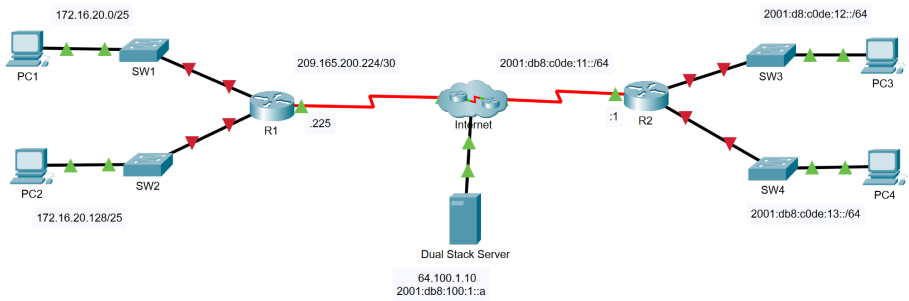
NETWORK LAB REPORT

BACKGROUND	
Date:	2026-02-15
Analyst:	BruceD
Lab Title:	Router Interface Provisioning: IPv4 and IPv6
Course Track:	CCNA SRWE
Environment:	Cisco Packet Tracer
Devices:	Routers R1, R2; Switches SW1–SW4; Hosts PC1–PC4; Dual Stack Server
Scope:	– Provision routed LAN interfaces with correct Layer-3 addressing. – Enable IPv6 forwarding on R2. – Validate reachability between LAN hosts and the dual-stack server.
ADDRESSING PLAN	
IPv4 LAN A:	R1 G0/0 = 172.16.20.1/25 PC1 = 172.16.20.10/25, GW 172.16.20.1
IPv4 LAN B:	R1 G0/1 = 172.16.20.129/25 PC2 = 172.16.20.138/25, GW 172.16.20.129
IPv4 WAN:	R1 S0/0/0 = 209.165.200.225/30 preconfigured
IPv4 Services Endpoint:	Server = 64.100.1.10 preconfigured
IPv6 LAN A:	R2 G0/0 = 2001:db8:c0de:12::1/64 PC3 = 2001:db8:c0de:12::a/64, GW fe80::2
IPv6 LAN B:	R2 G0/1 = 2001:db8:c0de:13::1/64 PC4 = 2001:db8:c0de:13::a/64, GW fe80::2
IPv6 WAN:	R2 S0/0/1 = 2001:db8:c0de:11::1/64 preconfigured
IPv6 Services Endpoint:	Server = 2001:db8:100:1::a preconfigured

BASELINE AND DESIGN NOTES

Network Intent and Routing Boundary

– R1 terminates two IPv4 LAN segments and provides default-gateway services.
– R2 terminates two IPv6 LAN segments and provides default-gateway services.
– WAN connectivity and the dual-stack server provide upstream validation targets.
– Static host addressing is used to keep test results deterministic.



Implementation Controls

– Subnet correctness: /25 masks prevent overlap and enforce correct default-gateway use.
– IPv6 prefixing: /64 supports standard Neighbor Discovery behavior on the LANs.
– Interface state: routed ports must be enabled to reach up/up.
– Forwarding: R2 requires `ipv6 unicast-routing` to route between IPv6 interfaces.
– Persistence: configurations are saved to startup configuration for reload safety.
– Verification: interface state checked; reachability confirmed with ICMP and ICMPv6.

CONFIGURATION

R1: IPv4 Interface Provisioning

Commands Applied

```
interface g0/0
   ip address 172.16.20.1 255.255.255.128
   no shutdown
interface g0/1
   ip address 172.16.20.129 255.255.255.128
   no shutdown
copy running-config startup-config
```

R2: IPv6 Interface Provisioning

Commands Applied

```
ipv6 unicast-routing
interface g0/0
   ipv6 address 2001:db8:c0de:12::1/64
   no shutdown
interface g0/1
   ipv6 address 2001:db8:c0de:13::1/64
   no shutdown
copy running-config startup-config
```

Interface State Confirmation (Post-Provisioning)

Interface status was reviewed after provisioning to confirm correct addressing and an operational `up/up` state.

```
R1#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     172.16.20.1     YES manual up
up
GigabitEthernet0/1     172.16.20.129   YES manual up
up
Serial0/0/0            209.165.200.225 YES manual up
up
Serial0/0/1            unassigned      YES unset
administratively down down
Vlan1                  unassigned      YES unset
administratively down down
R1#
```

```
R2#show ipv6 interface brief
GigabitEthernet0/0          [up/up]
    FE80::2
    2001:DB8:C0DE:12::1
GigabitEthernet0/1          [up/up]
    FE80::2
    2001:DB8:C0DE:13::1
Serial0/0/0                 [administratively down/down]
    unassigned
Serial0/0/1                 [up/up]
    FE80::207:ECFF:FE03:4702
    2001:DB8:C0DE:11::1
Vlan1                       [administratively down/down]
    unassigned
R2#
```

Host Provisioning

IPv4 Hosts

– PC1: `172.16.20.10/25`, gateway `172.16.20.1`.
– PC2: `172.16.20.138/25`, gateway `172.16.20.129`.

IPv6 Hosts

– PC3: `2001:db8:c0de:12::a/64`, gateway `fe80::2`.
– PC4: `2001:db8:c0de:13::a/64`, gateway `fe80::2`.
– Host link-local addresses are auto-generated and do not need to be fixed.

VALIDATION

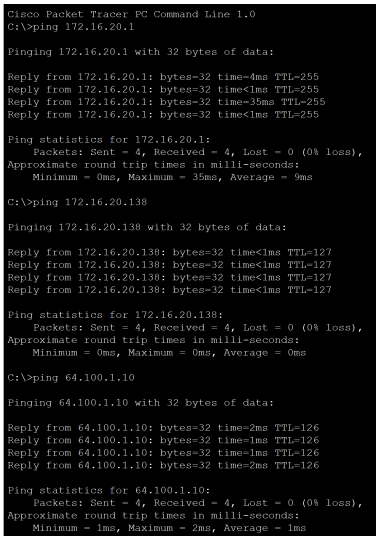
Verification Commands

– R1: `show ip interface brief` validates IPv4 addressing and state on G0/0 and G0/1.
– R2: `show ipv6 interface brief` validates IPv6 addressing and state on G0/0 and G0/1.
– Confirmation targets: correct interface mapping and correct prefix lengths.

IPv4 Reachability (ICMP Echo)

Local gateway
– PC1 to `172.16.20.1`: success (4/4).
Inter-LAN
– PC1 to `172.16.20.138`: success (4/4).
Upstream services
– PC1 to `64.100.1.10`: success (4/4).

ICMP echo was used to confirm gateway reachability, inter-LAN routing, and upstream server access.

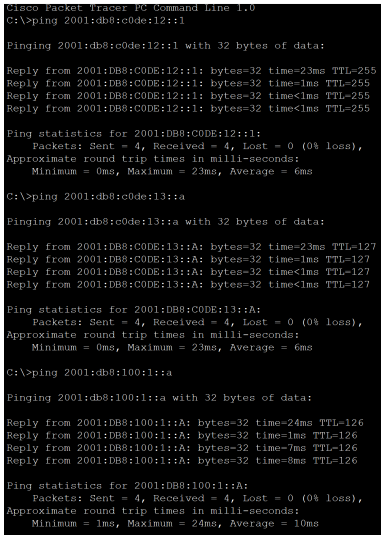


```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.20.1

Pinging 172.16.20.1 with 32 bytes of data:

Reply from 172.16.20.1: bytes=32 time=4ms TTL=255
Reply from 172.16.20.1: bytes=32 time<1ms TTL=255
Reply from 172.16.20.1: bytes=32 time=35ms TTL=255
Reply from 172.16.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 35ms, Average = 9ms

C:\>ping 172.16.20.138

Pinging 172.16.20.138 with 32 bytes of data:

Reply from 172.16.20.138: bytes=32 time<1ms TTL=127
Reply from 172.16.20.138: bytes=32 time<1ms TTL=127
Reply from 172.16.20.138: bytes=32 time<1ms TTL=127
Reply from 172.16.20.138: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.20.138:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 64.100.1.10

Pinging 64.100.1.10 with 32 bytes of data:

Reply from 64.100.1.10: bytes=32 time=2ms TTL=126
Reply from 64.100.1.10: bytes=32 time=1ms TTL=126
Reply from 64.100.1.10: bytes=32 time=1ms TTL=126
Reply from 64.100.1.10: bytes=32 time=2ms TTL=126

Ping statistics for 64.100.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

IPv6 Reachability (ICMPv6 Echo)

Local gateway
– PC3 to `2001:db8:c0de:12::1`: success (4/4).
Inter-LAN
– PC3 to `2001:db8:c0de:13::a`: success (4/4).
Upstream services
– PC3 to `2001:db8:100:1::a`: success (4/4).

ICMPv6 echo was used to confirm on-link resolution, routed reachability, and upstream server access.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:c0de:12::1

Pinging 2001:db8:c0de:12::1 with 32 bytes of data:

Reply from 2001:DB8:C0DE:12::1: bytes=32 time=23ms TTL=255
Reply from 2001:DB8:C0DE:12::1: bytes=32 time=1ms TTL=255
Reply from 2001:DB8:C0DE:12::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:C0DE:12::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DB8:C0DE:12::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 23ms, Average = 6ms

C:\>ping 2001:db8:c0de:13::a

Pinging 2001:db8:c0de:13::a with 32 bytes of data:

Reply from 2001:DB8:C0DE:13::A: bytes=32 time=23ms TTL=127
Reply from 2001:DB8:C0DE:13::A: bytes=32 time=1ms TTL=127
Reply from 2001:DB8:C0DE:13::A: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:C0DE:13::A: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:C0DE:13::A:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 23ms, Average = 6ms

C:\>ping 2001:db8:100:1::a

Pinging 2001:db8:100:1::a with 32 bytes of data:

Reply from 2001:DB8:100:1::A: bytes=32 time=24ms TTL=126
Reply from 2001:DB8:100:1::A: bytes=32 time=1ms TTL=126
Reply from 2001:DB8:100:1::A: bytes=32 time=7ms TTL=126
Reply from 2001:DB8:100:1::A: bytes=32 time=8ms TTL=126

Ping statistics for 2001:DB8:100:1::A:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 24ms, Average = 10ms
```

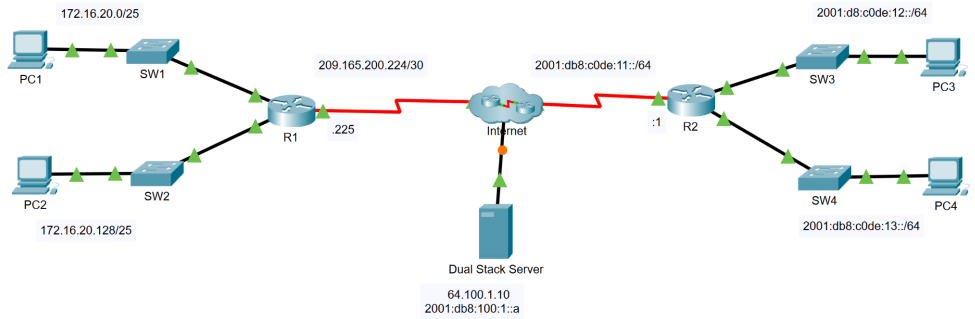
REPORT SUMMARY

Delivered State

– R1 provides IPv4 default gateways for two /25 LAN segments.
– R2 provides IPv6 default gateways for two /64 LAN segments and forwards IPv6 traffic.
– Host addressing matches the plan, with IPv6 link-local default gateway usage.
– End-to-end reachability to the dual-stack server is confirmed over IPv4 and IPv6.

Operational Snapshot

A topology snapshot was captured after provisioning to preserve the final layout used for validation.



Control Checklist

Check	State	Evidence
R1 G0/0 up/up with 172.16.20.1/25	✓	`show ip interface brief`
R1 G0/1 up/up with 172.16.20.129/25	✓	`show ip interface brief`
PC1/PC2 IPv4 and gateway correct	✓	host configuration
IPv4 reachability (GW, LAN, server)	✓	ICMP validation
R2 G0/0 up/up with 2001:db8:c0de:12::1/64	✓	`show ipv6 interface brief`
R2 G0/1 up/up with 2001:db8:c0de:13::1/64	✓	`show ipv6 interface brief`
PC3/PC4 IPv6 and gateway correct	✓	host configuration
IPv6 reachability (GW, LAN, server)	✓	ICMPv6 validation
Configuration saved to NVRAM	✓	`copy running-config startup-config`

Troubleshooting Notes

– If IPv4 pings fail: confirm correct /25 mask and correct gateway on PC1 and PC2.
– If IPv6 pings fail: confirm `ipv6 unicast-routing` on R2 and correct /64 prefixing.
– If an interface is down/down: verify cabling and apply `no shutdown`.
– If first ping fails but later succeeds: allow ARP and Neighbor Discovery to complete.